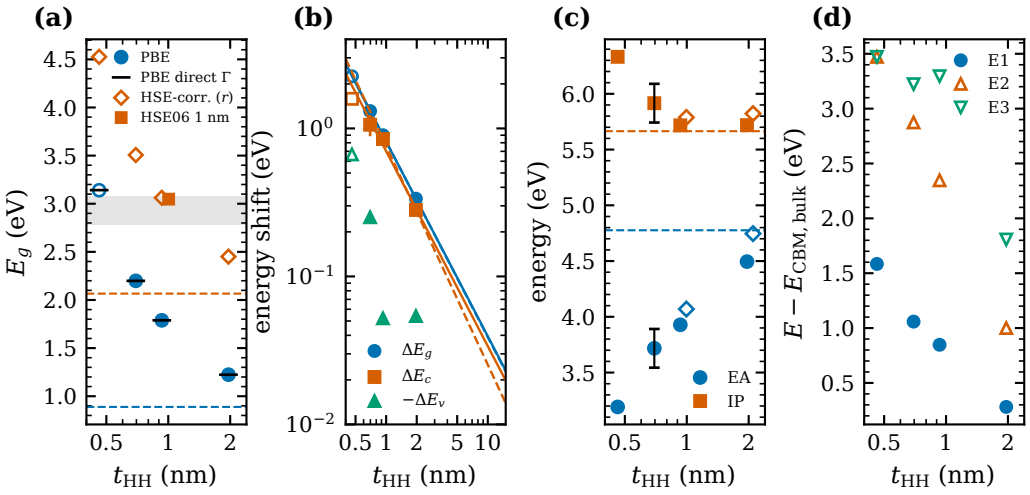
(a)
(b)
(c)
(d)
PBE
PBE direct Γ
HSE-corr. (r)
HSE06 1 nm
E_g (eV)
energy shift (eV)
ΔE_g
ΔE_c
$-\Delta E_v$
energy (eV)
EA
IP
$E - E_{\mathrm{CBM,\,bulk}}$ (eV)
E1
E2
E3
t_{HH} (nm)